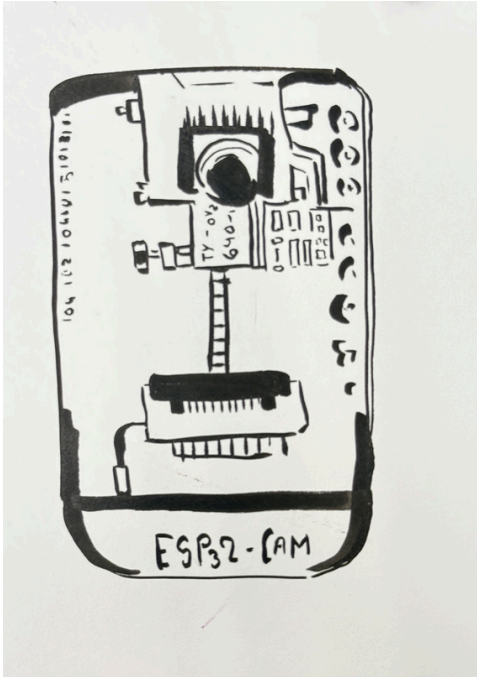
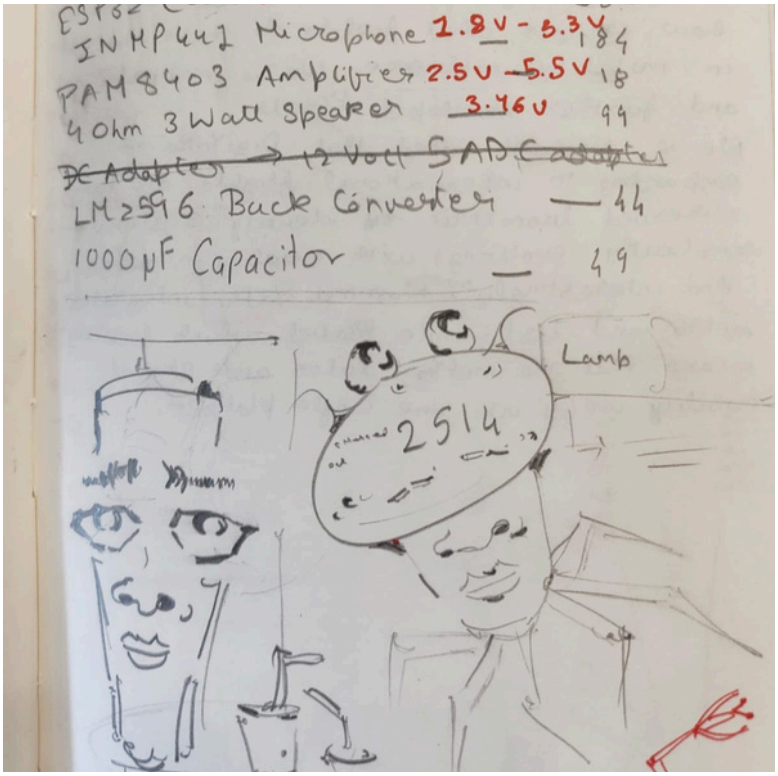
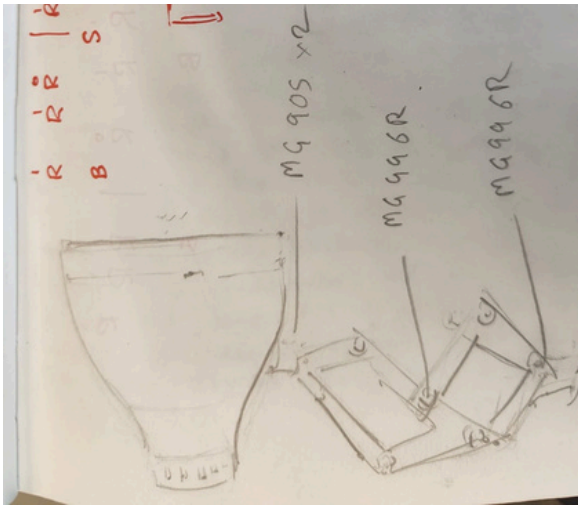
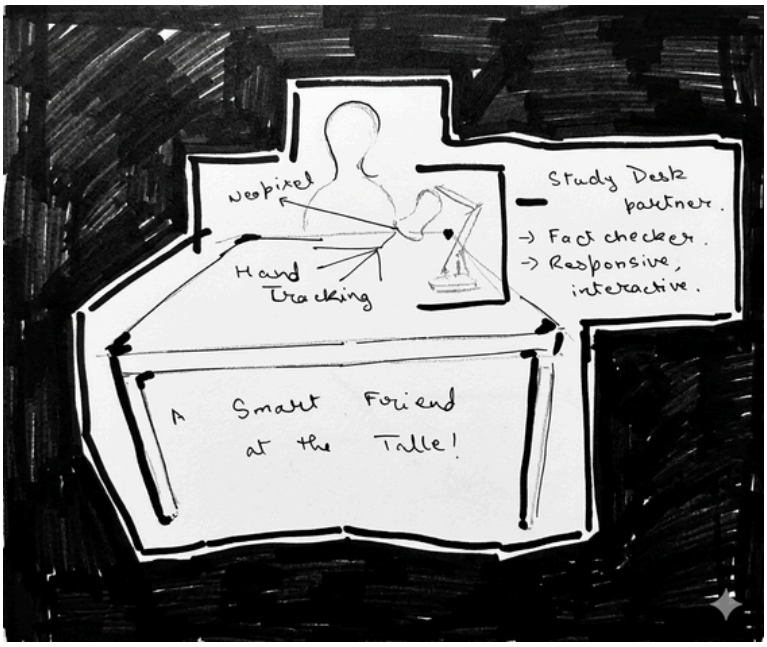
Week 1 – Process & Project

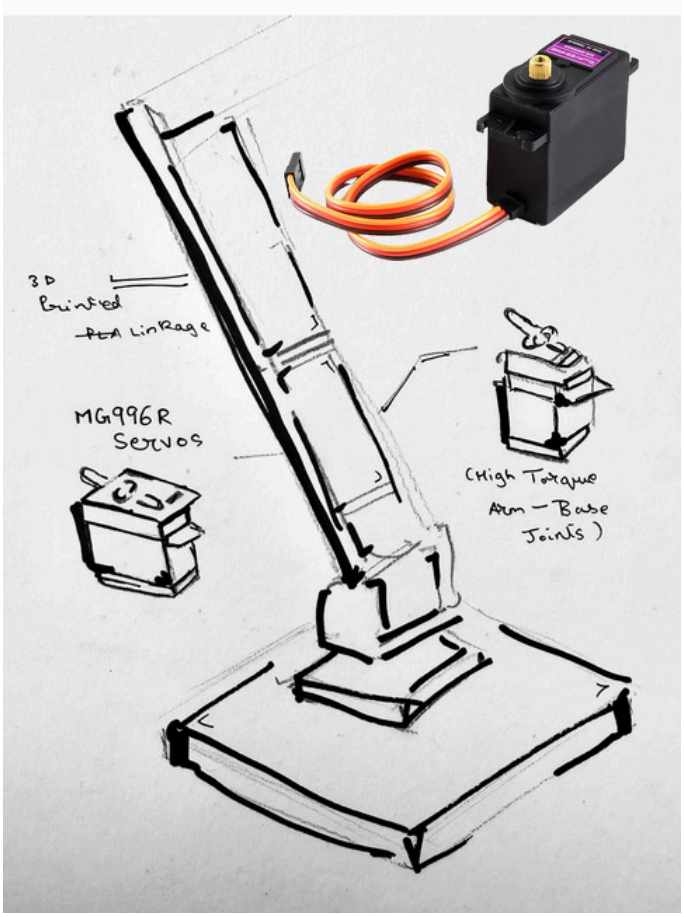
Brainstorming, ideation, **early prototyping**, distributing coding work, shortlisting components, understanding the power supply and distribution, calculations.

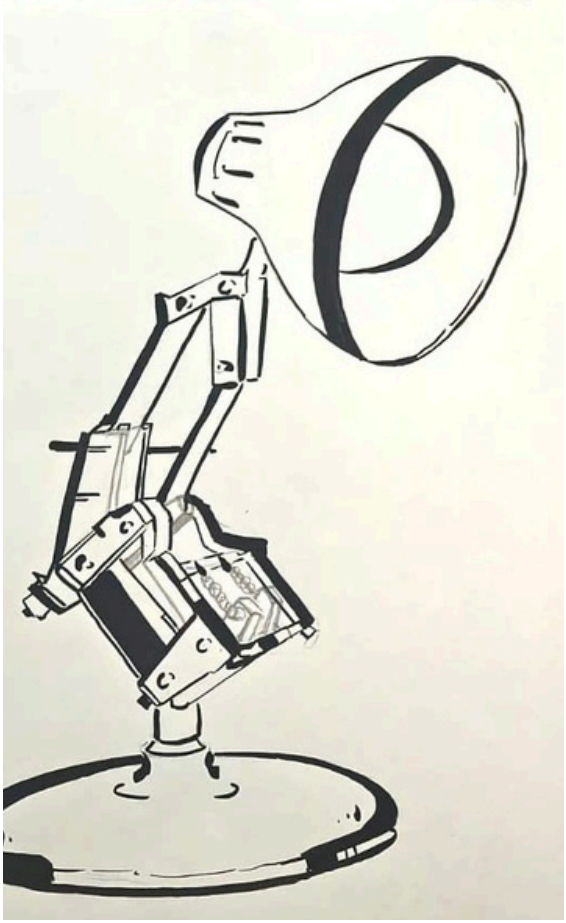


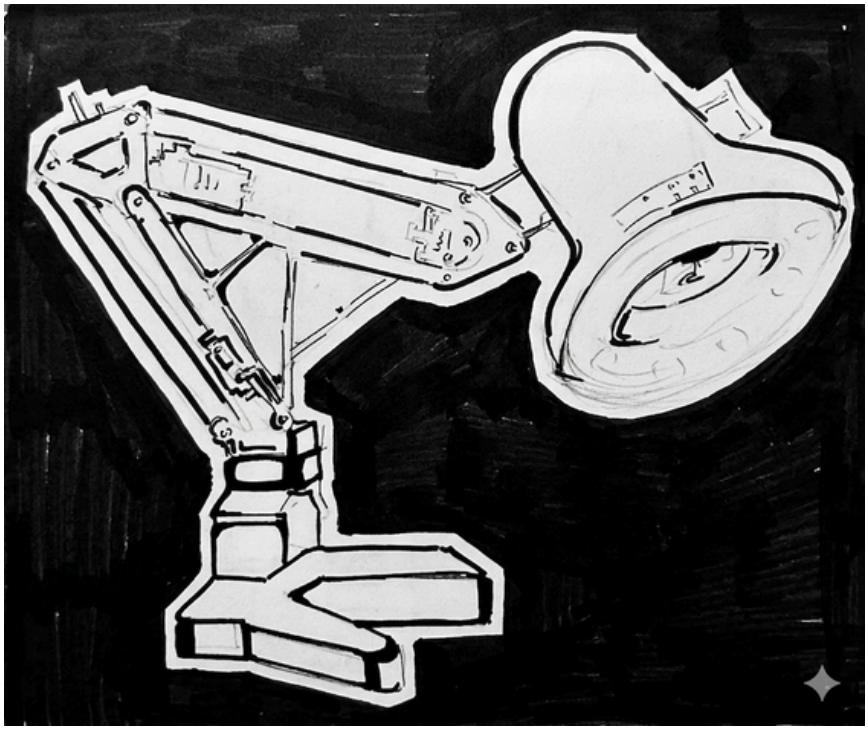


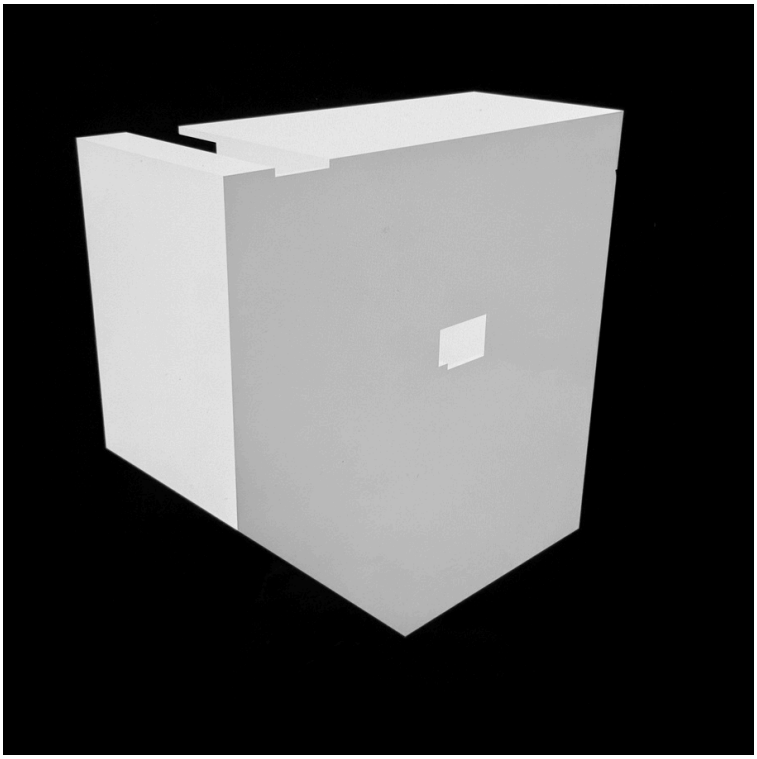



Neopixel
Hand Tracking
Study Desk partner.
-> Fact checker.
-> Responsive, interactive.
A Smart Friend at the Talle!

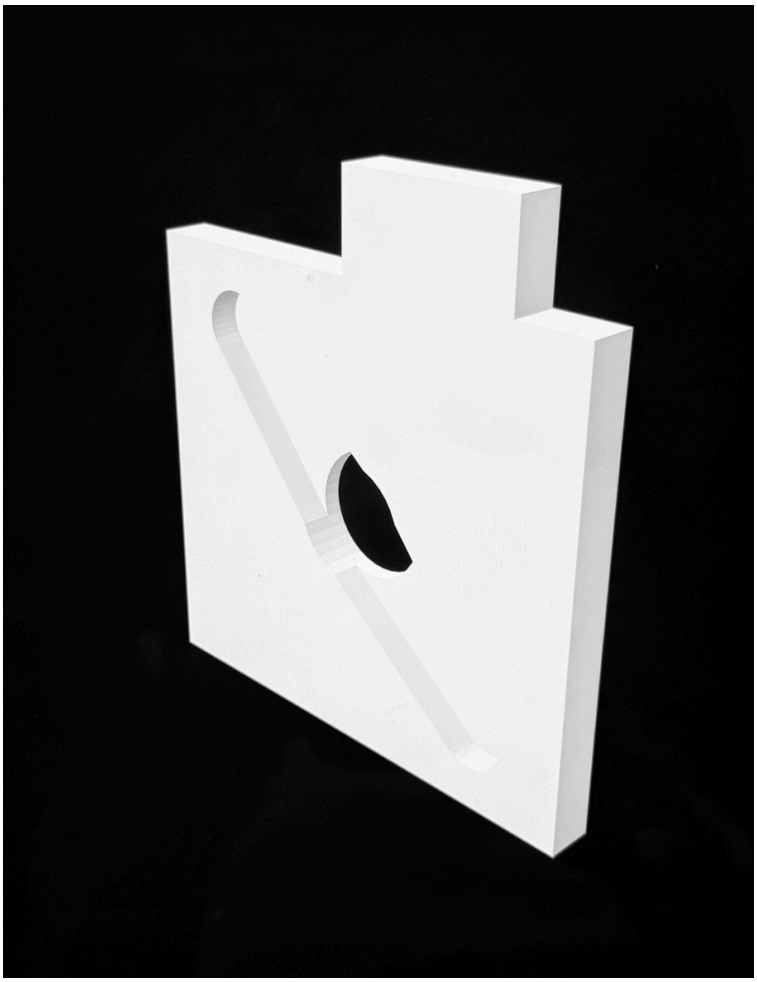

3D Printed
~~PLA~~ Linkage
MG996R Servos
(High Torque Arm - Base Joints)

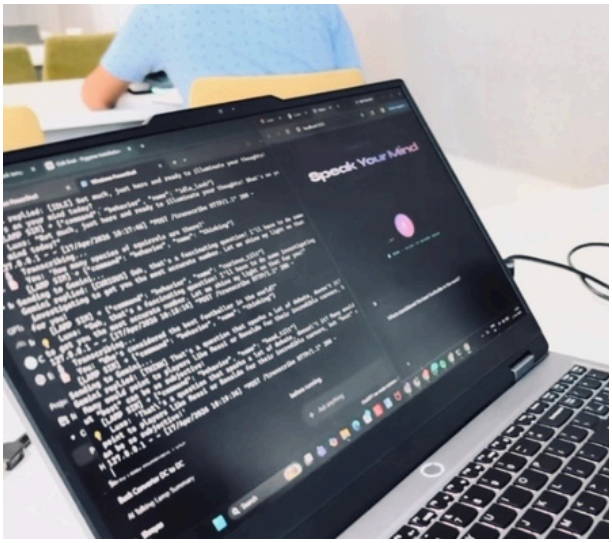






```
# Port 81 → WebSocket: servo behaviors +
NeoPixel commands
# Port 82 → WebSocket: PCM audio (when DAC
works)
# Idle animations run in background thread
# NeoPixel animations run in main loop
#

import machine, network, socket, time,
random, json, _thread, math, neopixel
from machine import Pin, PWM

#
#   CONFIG
#

WIFI_SSID     = "YOUR_WIFI_NAME"      # ←
your 2.4GHz network name
WIFI_PASSWORD = "YOUR_WIFI_PASSWORD"  # ←
your password

# NeoPixel
NEO_PIN   = 15        # ← GPIO15 (free pin,
away from servos/stepper)
NEO_COUNT = 12        # number of LEDs in
```





Testing early versions, Responive Gemini for our Lamp.